TUGAS PRAKTIKUM 3

Nama : Muhammad ‘Azmi Salam
NIM : 2406010
Kelas : C2

1. Implementasi RSA

```
def cari_fpb(a, b):
    """Mencari FPB (Faktor Persekutuan Terbesar)"""
    while b != 0:
        a, b = b, a % b
    return a

def cari_mod_inverse(e, phi):
    """Mencari Modular Multiplicative Inverse untuk Private Key (d)"""
    for d in range(1, phi):
        if (e * d) % phi == 1:
            return d
    return 1

def bangkitkan_kunci(p, q):
    # 1. Menghitung Modulus (n)
    n = p * q

    # 2. Menghitung Totient / Euler Phi
    phi = (p - 1) * (q - 1)

    # 3. Mencari Eksponen Publik (e)
    e = 2
    while e < phi:
        if cari_fpb(e, phi) == 1:
            break
        e += 1

    # 4. Mencari Eksponen Privat (d)
    d = cari_mod_inverse(e, phi)

    print("=" * 40)
    print("      PROSES PEMBANGKITAN KUNCI        ")
    print("=" * 40)
    print(f"Bilangan Prima p    : {p}")
    print(f"Bilangan Prima q    : {q}")
```

```python
    print(f"Modulus (n = p*q)    : {n}")
    print(f"Euler Phi (p-1)(q-1): {phi}")
    print("-" * 40)
    print(f"PUBLIC KEY (e, n)    : ({e}, {n})")
    print(f"PRIVATE KEY (d, n)   : ({d}, {n})")
    print("=" * 40)

    return {"n": n, "e": e, "d": d}

if __name__ == "__main__":
    # Gunakan bilangan prima kecil untuk belajar
    # Jangan gunakan angka yang sama!
    p = 71
    q = 89
    bangkitkan_kunci(p, q)
```

```
┌──(zicofarry㉿Asoes36)-[/mnt/d/Muhammad 'Azmi
└─$ python3 algo1.py
========================================
      PROSES PEMBANGKITAN KUNCI
========================================
Bilangan Prima p     : 71
Bilangan Prima q     : 89
Modulus (n = p*q)    : 6319
Euler Phi (p-1)(q-1): 6160
----------------------------------------
PUBLIC KEY (e, n)    : (3, 6319)
PRIVATE KEY (d, n)   : (4107, 6319)
========================================
```

```python
def cari_fpb(a, b):
    """Mencari FPB (Faktor Persekutuan Terbesar)"""
    while b != 0:
        a, b = b, a % b
    return a

def cari_mod_inverse(e, phi):
    """Mencari Modular Multiplicative Inverse untuk Private Key (d)"""
    for d in range(1, phi):
        if (e * d) % phi == 1:
            return d
    return 1
```

```python
def bangkitkan_kunci(p, q):
    # 1. Menghitung Modulus (n)
    n = p * q

    # 2. Menghitung Totient / Euler Phi
    phi = (p - 1) * (q - 1)

    # 3. Mencari Eksponen Publik (e)
    e = 2
    while e < phi:
        if cari_fpb(e, phi) == 1:
            break
        e += 1

    # 4. Mencari Eksponen Privat (d)
    d = cari_mod_inverse(e, phi)

    print("=" * 40)
    print("      PROSES PEMBANGKITAN KUNCI        ")
    print("=" * 40)
    print(f"Bilangan Prima p    : {p}")
    print(f"Bilangan Prima q    : {q}")
    print(f"Modulus (n = p*q)   : {n}")
    print(f"Euler Phi (p-1)(q-1): {phi}")
    print("-" * 40)
    print(f"PUBLIC KEY (e, n)   : ({e}, {n})")
    print(f"PRIVATE KEY (d, n)  : ({d}, {n})")
    print("=" * 40)

    return {"n": n, "e": e, "d": d}

def enkripsi(basis, eksponen, n):
    """Modular exponentiation: (basis ^ eksponen) mod n"""
    hasil = 1
    basis = basis % n
    while eksponen > 0:
        if eksponen % 2 == 1:
            hasil = (hasil * basis) % n
        basis = (basis * basis) % n
        eksponen = eksponen // 2
    return hasil
```

```python
if __name__ == "__main__":
    # Bangkitkan kunci
    kunci = bangkitkan_kunci(73, 97)
    print(f"Kunci E yang didapat: {kunci['e']}")

    pesan = 65

    # Enkripsi dengan kunci publik
    encrypted = enkripsi(pesan, kunci["e"], kunci["n"])
    print(f"Pesan asli     : {pesan}")
    print(f"Hasil Enkripsi: {encrypted}")

    # Dekripsi dengan kunci privat
    decrypted = enkripsi(encrypted, kunci["d"], kunci["n"])
    print(f"Hasil Dekripsi: {decrypted}")
    print(f"Pesan asli == Hasil Dekripsi? {pesan == decrypted}")
```

```
┌──(zicofarry㉿Asoes36)-[/mnt/d/Muhammad 'Azmi
└─$ python3 algo2.py
==========================================
      PROSES PEMBANGKITAN KUNCI
==========================================
Bilangan Prima p     : 73
Bilangan Prima q     : 97
Modulus (n = p*q)    : 7081
Euler Phi (p-1)(q-1): 6912
------------------------------------------
PUBLIC KEY (e, n)    : (5, 7081)
PRIVATE KEY (d, n)   : (2765, 7081)
==========================================
Kunci E yang didapat: 5
Pesan asli     : 65
Hasil Enkripsi: 5046
Hasil Dekripsi: 65
Pesan asli == Hasil Dekripsi? True
```

2. Implementasi Diffie-Hellman

```python
def diffie_hellman():
    # Parameter publik (disepakati bersama)
    p = 101  # bilangan prima
    g = 7    # generator

    # Kunci privat masing-masing (rahasia)
    a = 15   # kunci privat Alice
    b = 21   # kunci privat Bob

    # Kunci publik yang dikirimkan
    A = pow(g, a, p)   # Alice kirim ke Bob
    B = pow(g, b, p)   # Bob kirim ke Alice

    # Shared secret (dihitung masing-masing)
    secret_alice = pow(B, a, p)   # Alice hitung
    secret_bob   = pow(A, b, p)   # Bob hitung

    print("=" * 40)
    print("       DIFFIE-HELLMAN KEY EXCHANGE      ")
    print("=" * 40)
    print(f"Parameter publik: p={p}, g={g}")
    print(f"Kunci privat Alice (a): {a}")
    print(f"Kunci privat Bob   (b): {b}")
    print("-" * 40)
    print(f"Kunci publik Alice (A = g^a mod p): {A}")
    print(f"Kunci publik Bob   (B = g^b mod p): {B}")
    print("-" * 40)
    print(f"Shared secret Alice (B^a mod p): {secret_alice}")
    print(f"Shared secret Bob   (A^b mod p): {secret_bob}")
    print(f"Kunci sama? {secret_alice == secret_bob}")
    print("=" * 40)

if __name__ == "__main__":
    diffie_hellman()
```

```
┌──(zicofarry㉿Asoes36)-[/mnt/d/Muhammad '
└─$ python3 algo3.py
========================================
       DIFFIE-HELLMAN KEY EXCHANGE
========================================
Parameter publik: p=101, g=7
Kunci privat Alice (a): 15
Kunci privat Bob   (b): 21
----------------------------------------
Kunci publik Alice (A = g^a mod p): 39
Kunci publik Bob   (B = g^b mod p): 83
----------------------------------------
Shared secret Alice (B^a mod p): 39
Shared secret Bob   (A^b mod p): 39
Kunci sama? True
========================================
```

Penyadap tidak dapat menghitung shared secret karena di balik algoritma Diffie-Hellman terdapat persoalan matematis yang dinamakan Discrete Logarithm Problem (Masalah Logaritma Diskrit). Walaupun penyadap tahu $A = g^a \bmod p$, secara komputasi sangatlah sulit (bahkan mustahil dalam waktu wajar untuk bilangan prima yang besar) untuk membalikkan rumus tersebut agar bisa menemukan kunci privat asli (a atau b). Tanpa mengetahui minimal satu kunci privat (a atau b), penyadap tidak akan bisa merakit shared secret.

3. Praktikum TLS

Hasil curl -v http://localhost (HTTP tanpa TLS).

```
┌──(zicofarry㉿Asoes36)-[/mnt/d/Muhammad 'Azmi Salam/Kuliah/Semester 4/Praktikum-Kri
└─$ curl -v http://localhost
* Host localhost:80 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
*   Trying [::1]:80...
* connect to ::1 port 80 from ::1 port 45064 failed: Connection refused
*   Trying 127.0.0.1:80...
* Established connection to localhost (127.0.0.1 port 80) from 127.0.0.1 port 55192
* using HTTP/1.x
> GET / HTTP/1.1
> Host: localhost
> User-Agent: curl/8.18.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Server: nginx
< Date: Sun, 19 Apr 2026 14:48:13 GMT
< Content-Type: text/html
< Content-Length: 28
< Last-Modified: Sun, 19 Apr 2026 14:10:49 GMT
< Connection: keep-alive
< ETag: "69e4e269-1c"
< Accept-Ranges: bytes
<
<h1>Halo Praktikum TLS</h1>
* Connection #0 to host localhost:80 left intact
```

Hasil capture tcpdump pada port 80 (tunjukkan bahwa data terlihat dalam plain text).

```
< Last-Modified: Sun, 19 Apr 2026 14:10:49 GMT
< Connection: keep-alive
< ETag: "69e4e269-1c"
< Accept-Ranges: bytes
<
<h1>Halo Praktikum TLS</h1>
* Connection #0 to host localhost:80 left intact

┌──(zicofarry㉿Asoes36)-[/mnt/d/Muhammad 'Azmi Salam/Kuliah/Semester 4
/Praktikum-Kripto/Day-3]
└─$ curl http://localhost
<h1>Halo Praktikum TLS</h1>

┌──(zicofarry㉿Asoes36)-[/mnt/d/Muhammad 'Azmi Salam/Kuliah/Semester 4
/Praktikum-Kripto/Day-3]
└─$

┌──(zicofarry㉿Asoes36)-[/mnt/d/Muhammad 'Azmi Salam/Kuliah/Semester 4
/Praktikum-Kripto/Day-3]
└─$

┌──(zicofarry㉿Asoes36)-[/mnt/d/Muhammad 'Azmi Salam/Kuliah/Semester 4
/Praktikum-Kripto/Day-3]
└─$ curl http://localhost
<h1>Halo Praktikum TLS</h1>

┌──(zicofarry㉿Asoes36)-[/mnt/d/Muhammad 'Azmi Salam/Kuliah/Semester 4
/Praktikum-Kripto/Day-3]
└─$
```

```
┌──(Message from Kali developers)
│
│ This is a minimal installation of Kali Linux, you likely
│ want to install supplementary tools. Learn how:
│ ⇒ https://www.kali.org/docs/troubleshooting/common-minimum-setup/
└──(Run: "touch ~/.hushlogin" to hide this message)
┌──(zicofarry㉿Asoes36)-[/mnt/d/Muhammad 'Azmi Salam/Kuliah/Semester 4
/Praktikum-Kripto/Day-3]
└─$ sudo tcpdump -i any -A port 80
[sudo] password for zicofarry:
tcpdump: WARNING: any: That device doesn't support promiscuous mode
(Promiscuous mode not supported on the "any" device)
tcpdump: verbose output suppressed, use -v[v]... for full protocol dec
ode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot len
gth 262144 bytes
21:54:26.952164 lo    In  IP6 ip6-localhost.54966 > ip6-localhost.http
: Flags [S], seq 277807853, win 65476, options [mss 65476,sackOK,TS va
l 683534291 ecr 0,nop,wscale 7], length 0
`        ?..(.@.................................P.............0......
...
(...........
21:54:26.952177 lo    In  IP6 ip6-localhost.http > ip6-localhost.54966
: Flags [R.], seq 0, ack 277807854, win 0, length 0
`.UH...@.............................P..........P.......
21:54:26.952399 lo    In  IP localhost.41612 > localhost.http: Flags [
S], seq 1056912900, win 65495, options [mss 65495,sackOK,TS val 379390
5545 ecr 0,nop,wscale 7], length 0
```

```
┌──(zicofarry㉿Asoes36)-[/mnt/d/Muhammad 'Azmi Salam/Kuliah/Semester 4
/Praktikum-Kripto/Day-3]
└─$ sudo tcpdump -i any -A port 80
[sudo] password for zicofarry:
tcpdump: WARNING: any: That device doesn't support promiscuous mode
(Promiscuous mode not supported on the "any" device)
tcpdump: verbose output suppressed, use -v[v]... for full protocol dec
ode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot len
gth 262144 bytes
21:54:26.952164 lo    In  IP6 ip6-localhost.54966 > ip6-localhost.http
: Flags [S], seq 277807853, win 65476, options [mss 65476,sackOK,TS va
l 683534291 ecr 0,nop,wscale 7], length 0
`       ?..(.@.....................................P.............0......
...
(...........
21:54:26.952177 lo    In  IP6 ip6-localhost.http > ip6-localhost.54966
: Flags [R.], seq 0, ack 277807854, win 0, length 0
`.UH...@.................................P..........P.......
21:54:26.952399 lo    In  IP localhost.41612 > localhost.http: Flags [
S], seq 1056912900, win 65495, options [mss 65495,sackOK,TS val 379390
5545 ecr 0,nop,wscale 7], length 0
E..<.F@.@..s...........P>.6..........0.........
."g........
21:54:26.952410 lo    In  IP localhost.http > localhost.41612: Flags [
S.], seq 323238385, ack 1056912901, win 65483, options [mss 65495,sack
OK,TS val 3793905545 ecr 3793905545,nop,wscale 7], length 0
E..<..@.@.<..........P...D9.>.6......0.........
."g.."g.....
21:54:26.952423 lo    In  IP localhost.41612 > localhost.http: Flags [
.], ack 1, win 512, options [nop,nop,TS val 3793905545 ecr 3793905545]
, length 0
E..4.G@.@..z...........P>.6..D9......(.....
."g.."g.
```

```
r 3793905545], length 257: HTTP: HTTP/1.1 200 OK
E..5r=@.@............P...D9.>.6N.....).....
."g.."g.HTTP/1.1 200 OK
Server: nginx
Date: Sun, 19 Apr 2026 14:54:26 GMT
Content-Type: text/html
Content-Length: 28
Last-Modified: Sun, 19 Apr 2026 14:10:49 GMT
Connection: keep-alive
ETag: "69e4e269-1c"
Accept-Ranges: bytes

<h1>Halo Praktikum TLS</h1>

21:54:26.953181 lo    In  IP localhost.41612 > localhost.http: Flags [
.], ack 258, win 510, options [nop,nop,TS val 3793905546 ecr 379390554
6], length 0
E..4.I@.@..x...........P>.6N.D:......(.....
."g.."g.
21:54:26.953408 lo    In  IP localhost.41612 > localhost.http: Flags [
F.], seq 74, ack 258, win 512, options [nop,nop,TS val 3793905546 ecr
3793905546], length 0
E..4.J@.@..w...........P>.6N.D:......(.....
."g.."g.
21:54:26.953532 lo    In  IP localhost.http > localhost.41612: Flags [
F.], seq 258, ack 75, win 512, options [nop,nop,TS val 3793905546 ecr
3793905546], length 0
E..4r>@.@............P...D:.>.6O.....(.....
```

Hasil curl -vk https://localhost (HTTP dengan TLS).

```
┌──(zicofarry㉿Asoes36)-[/mnt/d/Muhammad 'Azmi Salam/Kuliah/Semester 4/Praktikum-Kripto/Day-3]
└─$ curl -vk https://localhost
* Host localhost:443 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
*   Trying [::1]:443...
* connect to ::1 port 443 from ::1 port 49980 failed: Connection refused
*   Trying 127.0.0.1:443...
* ALPN: curl offers h2,http/1.1
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
* SSL Trust: peer verification disabled
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.3 (IN), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (IN), TLS handshake, Encrypted Extensions (8):
* TLSv1.3 (IN), TLS handshake, Certificate (11):
* TLSv1.3 (IN), TLS handshake, CERT verify (15):
* TLSv1.3 (IN), TLS handshake, Finished (20):
* TLSv1.3 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (OUT), TLS handshake, Finished (20):
* SSL connection using TLSv1.3 / TLS_AES_256_GCM_SHA384 / X25519MLKEM768 / RSASSA-PSS
* ALPN: server accepted http/1.1
* Server certificate:
*   subject: C=AU; ST=Some-State; O=Internet Widgits Pty Ltd
*   start date: Apr 19 14:16:48 2026 GMT
*   expire date: Apr 19 14:16:48 2027 GMT
*   issuer: C=AU; ST=Some-State; O=Internet Widgits Pty Ltd
*   Certificate level 0: Public key type RSA (2048/112 Bits/secBits), signed using sha256WithRSAEncryption
*  SSL certificate verification failed, continuing anyway!
* Established connection to localhost (127.0.0.1 port 443) from 127.0.0.1 port 60360
* using HTTP/1.x
> GET / HTTP/1.1
> Host: localhost
> User-Agent: curl/8.18.0
> Accept: */*
```

```
> Accept: */*
>
* Request completely sent off
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
< HTTP/1.1 200 OK
< Server: nginx
< Date: Sun, 19 Apr 2026 14:56:21 GMT
< Content-Type: text/html
< Content-Length: 28
< Last-Modified: Sun, 19 Apr 2026 14:10:49 GMT
< Connection: keep-alive
< ETag: "69e4e269-1c"
< Accept-Ranges: bytes
<
<h1>Halo Praktikum TLS</h1>
* Connection #0 to host localhost:443 left intact

┌──(zicofarry㉿Asoes36)-[/mnt/d/Muhammad 'Azmi Salam/Kul
└─$
```

Hasil capture tcpdump pada port 443 (tunjukkan bahwa data terenkripsi).

```
┌──(zicofarry㉿Asoes36)-[/mnt/d/Muhammad 'Azmi Salam/Kuliah/Semester 4/Praktikum-Kripto/Day-3]
└─$ sudo tcpdump -i any -A port 443
tcpdump: WARNING: any: That device doesn't support promiscuous mode
(Promiscuous mode not supported on the "any" device)
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
21:58:39.432027 lo    In  IP6 ip6-localhost.52224 > ip6-localhost.https: Flags [S], seq 128238549, win 65476, options [mss 65476,sackO
K,TS val 683786789 ecr 0,nop,wscale 7], length 0
`.#..(.@.................................................0.........
(..%........
21:58:39.432035 lo    In  IP6 ip6-localhost.https > ip6-localhost.52224: Flags [R.], seq 0, ack 128238550, win 0, length 0
`..>...@.............................................P.......
21:58:39.432154 lo    In  IP localhost.40630 > localhost.https: Flags [S], seq 4080127189, win 65495, options [mss 65495,sackOK,TS val
 3794158043 ecr 0,nop,wscale 7], length 0
E..<p.@.@..7............1...........0.........
.&A.........
21:58:39.432163 lo    In  IP localhost.https > localhost.40630: Flags [S.], seq 1669036021, ack 4080127190, win 65483, options [mss 65
495,sackOK,TS val 3794158043 ecr 3794158043,nop,wscale 7], length 0
E..<..@.@.<............c{w..1.......0.........
.&A..&A.....
21:58:39.432171 lo    In  IP localhost.40630 > localhost.https: Flags [.], ack 1, win 512, options [nop,nop,TS val 3794158043 ecr 3794
158043], length 0
E..4p.@.@..>............1..c{w......(.....
.&A..&A.
21:58:39.434051 lo    In  IP localhost.40630 > localhost.https: Flags [P.], seq 1:1696, ack 1, win 512, options [nop,nop,TS val 379415
8045 ecr 3794158043], length 1695
E...p.@.@...............1..c{w...........
.&A..&A...........;....[O..}.a.        ..W.\..+...bP..o.. F..!....7.........qH.[.*.,..hq...........,.0...............].a.W.S...+./...
......\.`.V.R..........$.(.k.j.s.w.....m...#.'.g.@.r.v.....l...
...9.8......:...       ...3.2.....E.D...4...F.....Q.....P.....=...<...5.../...A.................     localhost........
...........................h2.http/1.1.........1.....<.:        .    ...................      .
...............................+........-.....3.......QI..x....Gg!.S....^e..D'"._.11.S{....\i.Q6..I....5..... *..L.2f..v..}....Y...
D.;.4f+..f..6"...Q..8..*.?\...!y.@,A.Q.UtRo........,.).\....
..b.)..k..i.A..8.
..d.9..$...]..]..@.i..V..%#.U7-t......q.S<...I7D.*.P.. |R..W.|.J..}.SM.....z.H.e...iH*y0...L.$..u8& i.....J...3.F......K.F..e@.....w..
..
..|..:..E.VC@7....,..\^.>.j....n.c.%G4.@Z@R.5...9.&...|.....rgX$9....O.{...J..GN.FF<i....8..b.U.AS..?.n.9...z.........q...S...(:..J.l.
FY.6.Q.9KEzx0.gD..K.G.hXL.k...Ai.!B}.-b..u.......dA..cK..[.}..P4lS.5t|.jQ&r....w".g..F.V.O{I.XuT..ll..52u....k......G.W.._!.6.`.YQm..h
...T  -.Q...G..T.A5.....y....qD.K.C........D..X|....7e...u'Q%f.......o0...'.Y..F.?...}...WjM#".........e8.[.[
.... +].4%..;k.G7/f.ka..%.......v...0qb V..5;5. O.UT..mc......I......L......L.ZI..Q.".T.'...w../b6~....d.k..nDe...$..] @].;...m..F9..C
...z?P...l~......../...........%.nv...67$...%..yu....'..D....'...p..E.....ssE.@..P.....<..z...".y.8.IU.F.s#..)Cqb..#[/y...G..g.d..V...
.a...cd.EE_~.....rQ.....<o...U.>...q.:........?.._:.S.zWl.rj...@...H.1....Jy'..3o..r....w4S.O.\..
"Ic......UR*..2N!.1g..........<Q%R.J...1..L.
..j<.n.Q..r=....r.S.LsR[....R23. EF I..H........|.*..+8..(w..xn."...T.........A`..=s...1.Y..8F1...../......C.._M..H.h.....P.E.@..XD...
m.... .=..B.h.......fTj..F-q...?...k,].........
21:58:39.434085 lo    In  IP localhost.https > localhost.40630: Flags [.], ack 1696, win 501, options [nop,nop,TS val 3794158045 ecr 3
794158045], length 0
E..4.W@.@.Fj............c{w..1.u.....(.....
.&A..&A.
21:58:39.437504 lo    In  IP localhost.https > localhost.40630: Flags [P.], seq 1:2493, ack 1696, win 512, options [nop,nop,TS val 379
4158048 ecr 3794158045], length 2492
E.      ..X@.@.<............c{w..1.u..........
.&A..&A............f../.nd....OkI...B.!f...V...@.       F..!....7.........qH.[.*.,..hq......n.+.....3.d...`.....'g.....I....b.....@B%
..... E....* ..W"...pmMk.L...5w.%..p..#l....(..XK.b..|.Y...[......W.K.....@.&...M.......U.u..{9.X$q2..w~..v9#\,.@OR.I>...6.d7.d.T*Y...
...^.qu.1/..uJ{!H. Ldx'Tw]Q.!....6p24..H....c...).........{\f...Ga........9.p.>....8.A1?bSU......G.\........Q...@9ae..q:...F[......s..
v.^.]3..,Pr.w.....w.!.=.........j.9....S.);e.w..rG...+..Q....I.,#...@q..kl..wY%..=.vR3O.Z...c....8W.    <........'m....@.....N.D.'X..#
.S..d.P...        .....2;F....A*oV........6...NP<.....\... .....`&5.UU..9..S'b....8Q..u.un...S...c0.L:T.i....t...4...../...3....$..>..
...:...a....8..b.....e....$*S..4.....7m....'....".>....xU...TS`.......XT.....%..      *..by.e.Us..R.W&ow...=....j.PD..{n.......2,.nj..
.......9.......8.....Z+...2.....-`....5..kI.RO*k.U..@{>...M..x!..y..$4#...!...E5..0#..........T..6E.D?.....PbQ>...........`~...^F..&.q
.&....r?U.H....1.Z.L.%%%..h.q..]+|.V..-....k.............0{......).!.IA.?.../.'7R..5.)..i&b.s...HZ.^.[g.(~.l.*g....,~Cm:...G...O/1}..
...).RcrlV...oL.....?...Bw.6@A.N{................W..E...........&.$...[....<......d......I.nw..''...P..X.]a,.+....).l.z.....2:.j....'.
#2..dT..?....8..?PH.#..W3.FG4+.6.t.g..........*._.....k.8..@9.u..z.3........B.._|{...~t>.....[>.yu...'..)..&.."q]...O..kmX....>...oCP.
.M....S@..c.X".<.......<..%....0l...dz.~~[e.*...#.f..z..s...e.... .m..F..   ..A
h.g.......$.xp.{...........&..#&..E...yx.,.w.\Q.:e.H.\....$.W.0y&}.X...;2.N.|.iX../|...'.qm....0.B.e....~9.....t....q..V.`......z.O.e
)*Ca."L..8...R..g(9.b..iS.V).w.X_...o..a\...>.....Vm..o.9$.rWW.......xsb,..v..*.VL.d.c.8..H~;N..a...O......-..ma.S\....     ....m.l.-5
.....i...+.x...
O.a..e].<A\..T.Q...C..<_..JG.....).M.......     ,F.>.F...N.kMM.PR.J1C8i2      ..}..."./.i}.70...3A_..>v...1.%.......4..g....>..ph....I
....r....RH....5.....3.....@...om.=.R..4<....s..e..u...,..].vzq-n....w'...c...Qu({...B..t/...M.fv0..P.6..O#wy.~K.R..g%f.Nt...;..6.....
`.......V.....{a...+......&E..3... .!I\..5?.!)t.....
.T......N(.k....].O....y.d.p...ty.i.:.sx......e..@D......K..U"R...'Gg..D.4....v0..'.......7...*.q.y:.3..N['.....w3...$...r....^.....O
R.&."....>.T.F%.........8.=...&....}.yx|..g.....%.]U.}Y. .Eq..v:=<.IR.X'...../Oj.^...}........i.x/._..%3..p(..l...TaZ..... 9^...jJ/..
..$....WS4.)..&..X....}~..(.iu.......l...M.....5.vh. ....:..v.qi~|......z......$..L..?..b)..I.`!.....S'..(.xW...(......R..].Jx.Ed..@UT
.J.j.p ......\{P.....i*.4.u4.+.......E\..J.....IF......k&.F.&..(S.M.........}8...%..._..r.5.n.B|Y.l...Q.S+1
21:58:39.437548 lo    In  IP localhost.40630 > localhost.https: Flags [.], ack 2493, win 498, options [nop,nop,TS val 3794158048 ecr 3
794158048], length 0
E..4p.@.@..<............1.uc{.......(.....
```

```
21:58:39.439425 lo    In  IP localhost.40630 > localhost.https: Flags [P.], seq 1696:1776, ack 2493, win 512, options [nop,nop,TS val
3794158050 ecr 3794158048], length 80
E...p.@.@................1.uc{.......x.....
.&A..&A..........E..b.0.. ..A......(.o.8"s.>.......6..|..E........cpM..N....|.7..P].%;.
21:58:39.439698 lo    In  IP localhost.40630 > localhost.https: Flags [P.], seq 1776:1871, ack 2493, win 512, options [nop,nop,TS val
3794158050 ecr 3794158048], length 95
E..S.Z@.@.EH............c{...1.$.....G.....
.&A..&A.........#...z<..%8.\.E......%..7..Q.._...p.,..n..Z....."....Cl..E.i#+.....O{T......c.dP...*$........"................XS|.g;
..-f.W*       ......R`J...A........^<B...TVN..k...K.()\.[.Q@  uP.[G-..my.\...-.....X7Y\.H...H.......2WCS~ZwX.`qA...).a.!....).m...O.P
=.....B...S......xK\..?.i.?.k...Y.
21:58:39.439827 lo    In  IP localhost.40630 > localhost.https: Flags [.], ack 3067, win 512, options [nop,nop,TS val 3794158051 ecr
794158051], length 0
E..4p.@.@..9............1.$c{.......(.....
.&A..&A.
21:58:39.439975 lo    In  IP localhost.https > localhost.40630: Flags [P.], seq 3067:3346, ack 1871, win 512, options [nop,nop,TS val
3794158051 ecr 3794158051], length 279
E..K.[@.@.EO............c{...1.$.....?.....
.&A..&A..... .....`...6.{..a....Vk......W. .{.T.....&.E...p.E..#....*t..R...B ....X...?.#.df..t.........b......G..kS...D....,..l....
.4......h....#....).u.......`....M_ZfN:1.....}..$......7.Nrw.N..W.So.....UQb.J1...Q..=.~d....vU.....~..Zf.,
..A*Q.9....K.^m.y..a...ed^W....i.#.V.Oz..:& 9
21:58:39.440226 lo    In  IP localhost.40630 > localhost.https: Flags [P.], seq 1871:1895, ack 3346, win 512, options [nop,nop,TS val
3794158051 ecr 3794158051], length 24
E..Lp.@.@.. ............1.$c{.......@.....
.&A..&A..........l`.7.9hl5../r.
21:58:39.440328 lo    In  IP localhost.https > localhost.40630: Flags [F.], seq 3346, ack 1895, win 512, options [nop,nop,TS val 3794
58051 ecr 3794158051], length 0
E..4.\@.@.Fe............c{...1.<.....(.....
.&A..&A.
21:58:39.440401 lo    In  IP localhost.40630 > localhost.https: Flags [F.], seq 1895, ack 3347, win 512, options [nop,nop,TS val 3794
58051 ecr 3794158051], length 0
E..4p.@.@..7............1.<c{.......(.....
.&A..&A.
21:58:39.440420 lo    In  IP localhost.https > localhost.40630: Flags [.], ack 1896, win 512, options [nop,nop,TS val 3794158051 ecr
794158051], length 0
E..4.]@.@.Fd............c{...1.=.....(.....
```

Pada HTTP biasa (port 80), data yang dikirimkan antara client dan server berbentuk plain text tanpa pelindung. Sehingga, ketika diamati melalui packet sniffer seperti tcpdump, seluruh isi data web dapat dibaca dengan sangat jelas. Sedangkan pada HTTPS (port 443 dengan TLS), seluruh pertukaran data telah dienkripsi secara asimetris/simetris. Saat disadap menggunakan tcpdump, data yang tertangkap hanyalah teks acak / karakter acak terenkripsi yang mustahil dibaca pesannya tanpa memiliki kunci privat terkait.

4. SHA-1

a. Apa itu fungsi hash kriptografi?

Fungsi hash kriptografi adalah sebuah fungsi matematis algoritme satu arah (one-way) yang menerima input (data / teks) berukuran berapa pun, dan memetakannya menjadi deretan output karakter / ukuran yang selalu tetap dan unik (disebut hash value atau digest). Jika satu huruf saja pada dokumen input berubah, maka seluruh output hash akan berubah secara drastis (Avalanche Effect).

b. Mengapa SHA-1 tidak lagi dianggap aman?

SHA-1 menghasilkan output sepanjang 160-bit. Algoritme ini tidak lagi dianggap aman sejak 2017 dikarenakan kerentanan secara komputasi terhadap serangan Collision (Tabrakan Hash). Collision attack berhasil didemonstrasikan ketika peneliti dari Google berhasil membuat dua buah dokumen (file PDF) yang isinya benar-benar berbeda, namun menghasilkan nilai hash SHA-1 yang persis sama (SHAttered). Hal ini mematahkan sifat "Collision Resistance" sebuah fungsi keamanan hash, sehingga integritas data tak lagi bisa dijamin bila memanfatkan SHA-1.

c. Contoh Penggunaan Fungsi Hash di kehidupan nyata:

1. Penyimpanan Password Database (Hashing Password): Saat kita mendaftar media sosial (Instagram, X, Facebook, dll), server tidak menyimpan password asli kita melainkan nilai hash dari password tersebut, sehingga administrator sekalipun tidak bisa mengintip password asli kita.
2. Verifikasi Integritas File Unduhan: Saat kita men-download file OS besar (misalnya file iso Linux atau Windows di internet) atau mendownload game dan software, pengembang biasanya menyediakan kode MD5 Hash atau SHA-256 Hash di situs webnya. Fungsi hash digunakan untuk memastikan file yang dikirim / di-download pengguna tersebut mulus 100% dan terbebas dari sisipan malware atau file korup, dengan membandingkan nilai hash dari hasil unduhan dengan originalitas dari server resminya.